OPERATING SYSTEMS (CS304)

Paging & Segmentation

Learning Objectives

Understand memory management, paging, segmentation, address translation, fragmentation, and their advantages and disadvantages.

Introduction to Memory Management

Memory management is a fundamental function of an operating system that controls and coordinates computer memory allocation among processes.

What is Paging?

Paging is a memory management technique that divides logical memory into fixed-size pages and physical memory into frames of the same size.

Paging Terminology

• Page: Fixed-size block of logical memory
• Frame: Fixed-size block of physical memory
• Page Table: Maps pages to frames
• Logical Address
• Physical Address

Address Translation in Paging

The CPU generates a logical address. The page number is used to access the page table and obtain the corresponding frame number. The frame number and offset form the physical address.

Advantages of Paging

• Eliminates external fragmentation
• Efficient memory utilization
• Supports virtual memory
• Simplifies memory allocation

Disadvantages of Paging

• Internal fragmentation may occur
• Additional memory required for page tables
• Address translation overhead

What is Segmentation?

Segmentation is a memory management technique that divides a program into variable-sized logical segments such as code, data, and stack.

Segment Table

The segment table stores the base address and limit of each segment. Logical addresses consist of a segment number and offset.

Advantages of Segmentation

• Supports logical program structure
• Easy sharing and protection
• Better program organization

Disadvantages of Segmentation

• External fragmentation
• Complex memory allocation
• Compaction may be required

Paging vs Segmentation

Paging uses fixed-size blocks, while Segmentation uses variable-size blocks.
Paging is transparent to users, whereas Segmentation reflects program structure.
Paging may cause internal fragmentation; Segmentation may cause external fragmentation.

Applications

Modern operating systems often combine paging and segmentation to achieve efficient memory management and protection.

Exam Important Questions

1. Define Paging.
2. Explain address translation in Paging.
3. Define Segmentation.
4. Differentiate Paging and Segmentation.
5. Discuss advantages and disadvantages of both techniques.

Summary

Paging and Segmentation are important memory management techniques. Paging improves memory utilization using fixed-size blocks, while Segmentation organizes memory according to logical program structure.